MIL-P-13692D
31 October 1979
SUPERSEDING
MIL-P-13692C
28 August 1974

MILITARY SPECIFICATION

PLATE, LITHOGRAPHIC PRINTING, METAL SENSITIZED, WITH PROCESSING CHEMICALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an additive type sensitized lithographic plate, and processing chemicals, for lithographic offset printing, including line work and up to and including a 300 line screened image.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-A-250/1	- Aluminum 1100, Plate and Sheet.
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.
QQ-S-781	- Strapping, Steel, and Seals.

FSC 3610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

UU-P-465	- Paper, Book.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-F-320	- Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

MILITARY

MIL-B-131	- Barrier Materials, Watervapor-proof, Flexible, Heat-Sealable.
MIL-I-3606/1	- Ink, Printing: Lithographic Offset, Black.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The sensitized plates shall consist of aluminum sheets with a light-sensitive coating which shall form a positive lithographic image when exposed to light through a negative and developed with the processing chemicals furnished.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:10Z
Check the source to verify that this is the current version before use.

3.2.1 Aluminum sheet. The aluminum sheet shall conform to QQ-A-250/1 or QQ-A-250/2. The temper shall be that normally used by the contractor. The thickness of the sheet shall be between 0.008 and 0.012 inch (0.2 and 0.3 millimeter (mm)). Variations in thickness within a given sheet shall not exceed 0.001 inch (0.03 mm).

3.3 Environmental conditions. The plates and chemicals shall not be damaged and the chemical containers shall not break under the following storage conditions:

High - plus 155° F (plus 68° C), plus or minus 2° F (1° C), for periods up to 4 hours daily for 2 days without benefit of solar radiation and with negligible air movement.
Low - minus 50° F (minus 46° C) for 12 hours without benefit of solar radiation and with negligible air movement.

3.4 Size. Unless otherwise specified (see 6.2), the size of plates shall be 27 by 30 inches, plus or minus 0.125 inch, (686 by 762 mm, plus or minus 3.18 mm).

3.5 Coating. The sensitized coating shall be uniform and free from coating streaks, pinholes, dirt, abrasion marks, spots, and mottling which will effect the printing quality of the plates. The plates shall be sensitized on both sides.

3.5.1 Characteristics of coating. When the coated plates are exposed for processing, the variation in sensitivity of coating between plates shall be not more than plus or minus 1 critical step using a 21-step sensitivity guide. The solid steps of the sensitivity guide shall be between Steps 6 to 8, and the number of gray steps following the critical step shall be not more than 4 after being rubbed back with a moist finger.

3.6 Processing. The processing of the plates shall be accomplished with not more than two chemical solutions. Chemical solutions for processing the plates shall be furnished in the quantities of 3 quarts (2.84 liters) of desensitizer and 3 pints (1.42 liters) of image developer per 25 plates.

3.7 Printing quality. The plates, containing a combination of solids, line, and 150 and 300 line halftone images, shall produce not less than 20,000 impressions per plate. The printed sheet shall have a fine highlight dot and a clean, open-shadow dot on both the 150 line

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:10Z
Check the source to verify that this is the current version before use.

and 300 line halftone images. A comparison of the first and last sheets under a 9-power magnifying glass shall show no deterioration of the image or evidence of scum in the nonprinting areas.

3.8 Expiration date. The expiration date of the sensitized plates shall be not less than 1 year from the date of acceptance.

3.9 Labeling of chemical containers. Each chemical container shall have a label affixed thereto containing the following information:

(a) Title of chemical.
(b) Date of manufacture.
(c) Manufacturer's name and trade mark.
(d) Number or coding to indicate the corresponding plates for which the chemicals are used.

3.10 Instruction sheet. An instruction sheet giving complete details regarding the handling of the plates during exposure, processing, and printing shall be furnished in each package of 25 plates.

3.11 Workmanship. The sensitized plates shall be free from dents, kinks, and waves. The edges of the plates shall be straight, the corners shall be square and true, and all burrs and rough edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.1.1 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects.

4.3.1.2 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 4 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. Plates selected for examination in accordance with 4.3.1 shall be examined for the following defects:

101. Size of plates not as specified.
102. Thickness of plates not as specified.
103. Coating of the plate not as specified.
104. Labeling of containers missing, incomplete, or illegible.
105. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Test conditions. Unless otherwise specified herein, tests shall be performed at the climatic conditions existing at the place of test.

4.4.2.2 Storage temperature. The wrapped plates and chemicals preserved in accordance with 5.1 shall be placed in a controlled temperature chamber. The wrapped plates and chemicals shall then be subjected to the high and low temperatures as specified in 3.3. At the end of the temperature test the plates and chemicals shall be returned to normal room temperature and examined. Evidence of damage or breakage shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

4.4.2.3 Processing. The plate shall be exposed in contact with a test layout to produce a test press sheet conforming to Figure 1. After exposure, the plate shall be processed in accordance with the instruction sheet and using chemical solutions furnished with the plate. The plate shall then be examined. Nonconformance to 3.5 or 3.6 shall constitute failure of this test.

4.4.2.4 Printing quality. The plate shall be placed on an offset press and not less than 20,000 impressions shall be printed. Black ink for printing shall conform to MIL-I-3606/1, and paper shall conform to UU-P-465, Type II. Nonconformance to 3.7 shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials or containers not as specified for Level A or Level B. Each incorrect material or container shall be considered one defect.
107.	Plates not stacked in quantities of 25 for Level A.
108.	Sensitized surfaces of each plate not protected as specified for Level A.
109.	Fiberboard pad not placed on the top and bottom of the stack and secured as specified for Level A.
110.	Instruction sheet not packaged with each stack of plates for Level A.
111.	Plates not preserved in heat sealed bags as specified for Level A.
112.	Chemicals not preserved as specified for Level A.
113.	Bags of plates and boxes of chemicals not packed together in quantities for Level A or Level B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

114. Blocking not adequate to prevent free movement of contents within the box for Level A or Level B.
115. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Plates. The plates shall be uniformly stacked in quantities of 25. The sensitized surfaces of each plate shall be protected with the contractor's commercial interleaving paper that will not absorb moisture or with a sheet of plastic film conforming to L-P-378, Type I, Class 1, Grade A, Finish 1, thickness 0.0020. A fiberboard pad of the same length and width dimensions as the stacked plates and conforming to PPP-F-320, Type CF, Class Weather-Resistant, Variety SW, Grade W5c, shall be placed on the top and bottom of the stack. The pads shall be secured by no less than two girthwise and two lengthwise bands of tape conforming to PPP-T-45, Type III, Grade C. The stack of plates and one instruction sheet shall then be enclosed in a close-fitting heat-sealed bag fabricated from material conforming to MIL-B-131, Class 1.

5.1.1.2 Chemicals. The chemicals required for each 25 plates contained in the contractor's commercial container shall be preserved in a close-fitting box conforming to PPP-B-636, W6c. The chemicals shall be cushioned within the box to prevent free movement and damage. The box shall be closed and sealed as specified for Method V in the appendix to the box specification.

5.1.2 Commercial. The plates in quantities of 25, the chemicals required, and the instruction sheet, shall be preserved together in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. Four bags of plates and four boxes of chemicals shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. The components shall be blocked to prevent free movement within the box. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A.

5.2.2 Level B. Plates and chemicals shall be packed together as specified in 5.2.1 for Level A, except the box shall be Domestic Type, and strapping shall be Finish A.

5.2.3 Commercial. The preserved plates and preserved chemicals, in the quantity required, shall be packed together in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The presensitized plates are intended for use in printing line and halftone subjects on offset presses.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of plate required if other than as specified (see 3.4).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When strapping shall be Finish A (see 5.2.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

Custodians:

Army - ME
Air Force - 99

Review activity:

Army - GL
DLA - GS

User activity:

Navy - MC

Preparing activity:

Army - ME

Project 3610-0307

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

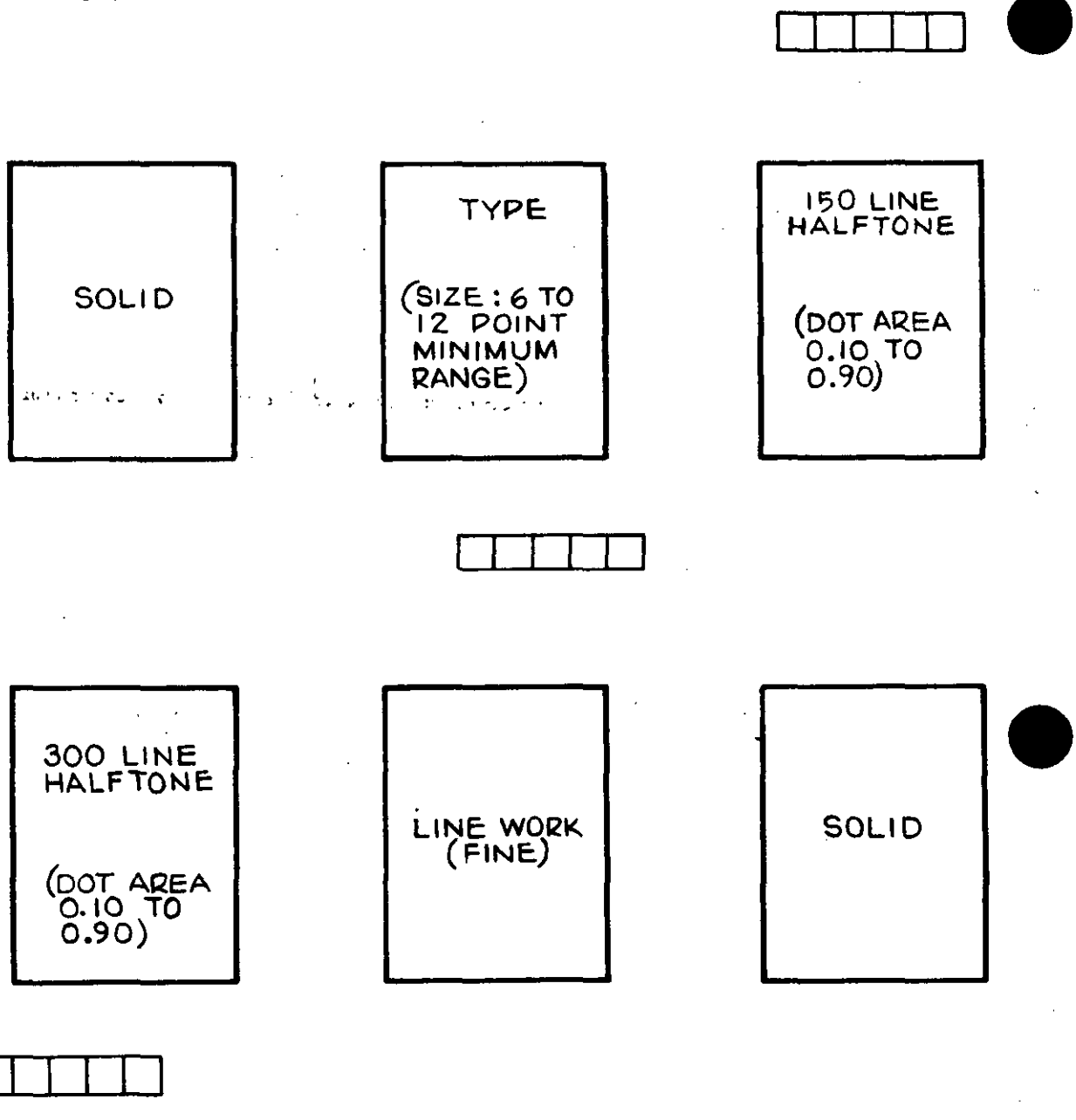


FIGURE 1. Test layout

X416

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on *lines on reverse side, staple* in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an *envelope* addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-P-13692D Plate, Lithographic Printing, Metal Sensitized, with Processing Chemicals

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:10Z
Check the source to verify that this is the current version before use.